

Quantifying Interlacing & Weaving

The techniques used by birds to construct nests may be partly reflected in the degree to which individual strands of nest material are interlaced. Quantifying these interlacing patterns could provide valuable insights into nest-building behaviour and construction rules.

One approach would be to infer interlacing structures from patterns of light and shadow in high-resolution images, enabling the reconstruction of strand overlap and weave topology (Li et al., 2020).

Alternatively (and probably more easily), when birds are provided with nest materials of different colours, image-analysis methods could be used to quantify the extent to which colours are intermixed throughout the nest. Such measures could serve as proxies for weaving complexity, material manipulation, and construction strategy.

Previous work has demonstrated that image-based analyses can successfully characterize weave textures and patterns in woven structures, including weaverbird nests (Bailey et al., 2015) and textiles (Li et al., 2020), suggesting that similar approaches could be adapted to quantify variation in nest interlacing and material integration.

Relevant references:

- Bailey IE, Backes A, Walsh PT, et al (2015) Image analysis of weaverbird nests reveals signature weave textures. R Soc Open Sci 2:. https://doi.org/10.1098/rsos.150074,
- Li, J., Wang, W., Deng, N., & Xin, B. (2020). A novel digital method for weave pattern recognition based on photometric differential analysis. Measurement., 152. https://doi.org/10.1016/j.measurement.2019.107336

Example Nests



Almost no diversity of colours
> only same-colour contact



Colours grouped together
> little inter-colour contact

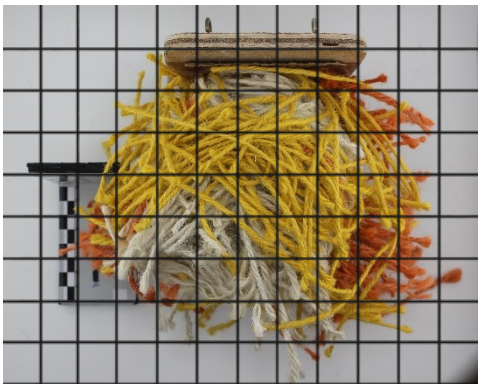


Colours somewhat mixed together
> medium inter-colour contact



Colours very mixed together
> high inter-colour contact

Potential approaches



1) **Colour mixing Index [easy]**

Divide picture into cells and determine number of colours by cell, then use a diversity index (e.g shannon's, spatial entropy index?). A nest with large segregated patches of colours would have low mixing values (of course need to account for the number of colours that have been used by this pair)

2) **Strand-by-strand neighbour analysis [somewhat easy]**

Each strand of material is segmented and treated as an object. The frequency with which different colours touch each other can be quantified (ratio of hetero-colour contacts over total contacts)

3) Spatial autocorrelation analysis [unclear difficulty]

[?] Metrics such as Moran's I or semivariograms can quantify the spatial scale over which colours remain clustered. High positive autocorrelation indicates large colour blocks, rapid decay in autocorrelation indicates fine-scale mixing.

Could use literature from landscape ecology ad treat pieces of string as “neighbouring patches”

4) Multi-scale texture analysis [difficult]

See method used by Bailey et al. (2015) to quantify regularity and orientation of material (in our case + colour). Texture descriptors calculated separately for each colour channel (grey-level co-occrence matric, wavelet decomposition, fourier texture analysis)

5) Interlacing reconstruction from occlusion and shadows [very difficult}

See method used by Li et al. (2020): using edge continuity, shadow patterns and brightness differences image processing identifies intersections and estimates which strand lies above the other.

This approach provides more information. This would also inform on the “weaviness” of the building (one strand is used to go under and over many strands vs. strands are deposited one by the other without much effort to interlace them into each other), actively providing information on construction behaviour. But we are not dealing with continuous strands of fibre like Li et al. we have short standard pieces of string so need to account for the fact that we have endings.

Example of Nests from the “rainbow experiment”

